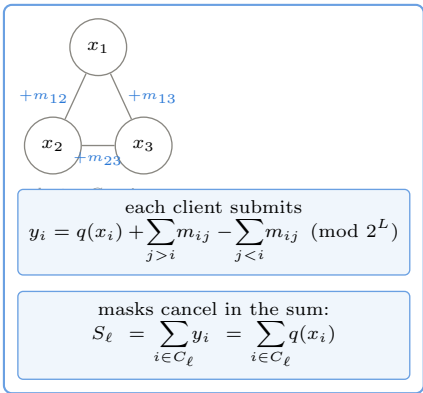


One-time setup (code-based HQC KEM): $(pk_i, sk_i) \leftarrow \text{KeyGen}$; for each pair (i, j) : $(c_{ij}, ss_{ij}) \leftarrow \text{Encaps}(pk_j)$; per-round masks $m_{ij}^{(r)} = G(\text{HKDF}(ss_{ij}, r))$ are reused every round (KEM cost paid once).

Confidentiality layer (code-based PQC)



k sums

Server sees *only*
cluster sums $S_1, \dots, S_k$
anonymity set = c ;
 $\Pr[S_\ell \text{ clean}] = (1 - f)^c$

Byzantine-robustness stage (CB-SAFE+)

flag cluster ℓ : $g_\ell^{(r)} = \mathbb{1}[\Delta \text{loss}(S_\ell) > 0]$
trust-free: $g_\ell^{(r)} = \mathbb{1}[\cos(S_\ell, \bar{S}) < 0]$

suspicion (re-randomized rounds):
 $s_i = |\{r : i \in \text{flagged}\}| / |\{r : i \text{ active}\}|$

adaptive exclusion: drop the high-suspicion minority (gap split), no fixed τ

robust aggregate & update:
 $w \leftarrow w + \text{RAgg}\{S_\ell : \ell \text{ clean}\}$

Laundering (why flat rules fail): a poisoned cluster mean $\mu = \frac{c - m(1+\gamma)}{c} h$ has $|\mu| = |h|$ at $\gamma^* = \frac{2c}{m} - 1$: same size, wrong direction.

broadcast w to clients (next round)